

Optimal Loading of Flatcars

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Abstract

We formally state the given problem, that of loading two flatcars with a subset of specified set of crates so as to minimize wasted floor space, subject to given constraints on size and weight. We find the minimum amount of wasted floor space in each of two possible interpretations of the given "local trucking" constraint, prove that these are optimal, and display all loadings that achieve these minima. A computer program is presented that finds optimal and near-optimal loadings. We discuss the computational complexity of a reasonable generalization of the problem, noting that our methods do not generalize to problems with greater numbers of parameters. We prove that the natural generalization is NP-complete and suggest a heuristic algorithm for it.

1. Introduction

We first restate the problem formally. The goal is to place packing crates on two railroad flatcars subject to constraints on thickness and weight. There are seven kinds of crates available in given numbers. Each kind has a particular thickness and weight but has the same width and height as all the others. Let n , w and t be vectors whose i^{th} component ($1 \leq i \leq 7$) is, respectively, the number of crates of type i available, those crates' weight in metric tons, and their thickness in centimeters. (The values of n_i , t_i and w_i are given in Table 1.) A particular loading of the flatcars is represented by two vectors N^1 and N^2 , where N_i^j , the i^{th} component of N^j , specifies the number of crates of type i on flatcar number j . The vectors N^j must satisfy the following constraints.

C1. We may not use more crates of each type than are available:
 $N_i^1 + N_i^2 \leq n_i$ for $1 \leq i \leq 7$.

Table 1.
Values of n_i , t_i , and w_i for $1 \leq i \leq 7$.

	Crate type (i)						
	1	2	3	4	5	6	7
n_i	8	7	9	6	6	4	8
t_i	48.7	52.0	61.3	72.0	48.7	52.0	64.0
w_i	2	3	1	0.5	4	2	1
							centimeters metric tons

C2. The total thickness of crates on each flatcar must not exceed 1020 centimeters: $N^j \cdot t \leq 1020$ for $j = 1, 2$. (Here " $\cdot$ " represents the dot product.)

C3. The total weight of crates on each flatcar must not exceed 40 metric tons: $N^j \cdot w \leq 40$ for $j = 1, 2$.

C4. Due to a "local trucking" restriction, the total thickness of crates of types 5, 6 and 7 must not exceed 302.7 cm. We do not know whether this limit applies individually to each flatcar or to the sum of both. We therefore solve twice for two different constraints. Let t' be a vector such that

$$t'_i = \begin{cases} 0 & 1 \leq i \leq 4 \\ t_i & 5 \leq i \leq 7. \end{cases}$$

Then either

$$N^1 \cdot t' + N^2 \cdot t' \leq 302.7 \quad (1)$$

or

$$N^j \cdot t' \leq 302.7 \quad \text{for } j = 1, 2. \quad (2)$$

We seek to minimize the total wasted floor space, which is equivalent to maximizing the total used floor space $N^1 \cdot t + N^2 \cdot t$.

It is unclear which interpretation of the local trucking constraint was intended. The constraint might apply to both cars if they share a single destination and are going to be unloaded onto a single truck. If, however, the cars are ultimately to separate, or if they must be loaded onto different trucks, then the constraint would apply to both.

2. Optimal Solutions

We present here a solution for each interpretation of the problem. For each, we give a loading that achieves the minimum value of wasted space and prove that this is the minimum value. The

complete set of loadings producing minimum wasted space we defer until the next section.

Theorem 1. Let $\mathcal{N}$ be the set of unordered pairs $\{N^1, N^2\}$ satisfying constraints C1, C3, and (1). (This is the set of admissible loadings under the "total length on both" interpretation of the local trucking restriction.) Then there exists a loading $\{N^1, N^2\} \in \mathcal{N}$ such that the total used space is 2039.4 centimeters, and this is the maximum amount of used space among loadings in $\mathcal{N}$. (This is just 0.6 cm short of the total available space on both flat cars.)

Proof: We consider the following loading:

$$N^1 = (4 \ 7 \ 4 \ 3 \ 0 \ 0 \ 0), \quad N^2 = (4 \ 0 \ 5 \ 3 \ 3 \ 3 \ 0).$$

We can easily check that the N^j satisfy the given constraints:

1. This loading satisfies the first constraint, since

$$N^1 + N^2 = (8 \ 7 \ 9 \ 6 \ 3 \ 3 \ 0) \quad \text{and} \quad n = (8 \ 7 \ 9 \ 6 \ 6 \ 4 \ 8),$$

so that $N_i^1 + N_i^2 \leq n_i$.

2. The loading does not exceed the total thickness limits, since

$$N^1 \cdot t = 1020 \leq 1020 \quad \text{and} \quad N^2 \cdot t = 1019.4 \leq 1020.$$

3. The loading does not exceed the total weight limits, since

$$N^1 \cdot w = 35.5 \leq 40 \quad \text{and} \quad N^2 \cdot w = 33.5 \leq 40.$$

4. We satisfy the local trucking constraint (1), since

$$N^1 \cdot t' + N^2 \cdot t' = 0 + 302.1 \leq 302.7.$$

Hence we have $\{N^1, N^2\} \in \mathcal{N}$. Also, $N^1 \cdot t + N^2 \cdot t = 1020 + 1019.4 = 2039.4$, so this loading does indeed achieve a wastage of only 0.6 cm.

Now, suppose $\{M^1, M^2\} \in \mathcal{N}$ and $M^1 \cdot t + M^2 \cdot t \geq 2039.4$. We will show that $M^1 \cdot t + M^2 \cdot t = 2039.4$.

First, we prove that $M_i^1 + M_i^2 = n_i$ for $1 \leq i \leq 4$. Suppose there is an i' , $1 \leq i' \leq 4$, such that $M_{i'}^1 + M_{i'}^2 < n_{i'}$. Then

$$\begin{aligned} M^1 \cdot t + M^2 \cdot t &= \sum_{i=1}^4 (M_i^1 + M_i^2) t_i + \sum_{i=5}^7 (M_i^1 + M_i^2) t_i \\ &\leq \left(\sum_{i=1}^4 (M_i^1 + M_i^2) t_i \right) + 302.7 \quad \text{by (1)} \\ &\leq \left(\sum_{i=1}^4 n_i t_i \right) - t_{i'} + 302.7. \end{aligned}$$

We compute that $\sum_{i=1}^4 n_i t_i = 1737.3$. Hence, since $\min_{i=1}^4 t_i = 48.7$,

$$M^1 \cdot t + M^2 \cdot t \leq 1737.3 - 48.7 + 302.7 = 2040 - 48.7.$$

Clearly, the loading $\{M^1, M^2\}$ must waste at least 48.7 centimeters, a contradiction.

Since $M_i^1 + M_i^2 = n_i$ for $1 \leq i \leq 4$,

$$M^1 \cdot t + M^2 \cdot t = 1737.3 + \sum_{i=5}^7 (M_i^1 + M_i^2) t_i.$$

We now assert that it is impossible to choose the numbers $(M_i^1 + M_i^2)$ for $5 \leq i \leq 7$ such that

$$302.1 < \sum_{i=5}^7 (M_i^1 + M_i^2) t_i \leq 302.7, \quad (3)$$

where 302.1 is the value attained by $M^j = N^j$. It will follow from this that $\{N^1, N^2\}$ achieves the maximum of used space on N .

Let $M_i = M_i^1 + M_i^2$ for $5 \leq i \leq 7$ and suppose that (3) holds. Then, since $7t_5 = 340.9 > 302.7$, $M_5 < 7$. We consider the possible M_5 :

- $M_5 = 0$

Since $t_6 = 52$ and $t_7 = 64$ are integers, the sum $\sum_{i=5}^7 M_i t_i$ will be an integer. Clearly (3) cannot then be satisfied, since there is no integer between 302.1 and 302.7.

- $M_5 = 1$

In this case, $\sum_{i=5}^7 M_i t_i = 48.7 + M_6 t_6 + M_7 t_7$. Since t_6 and t_7 are integers, the sum is now equal to an integer plus 0.7. Hence if (3) is satisfied, the sum equals 302.7. In that case, $M_6 t_6 + M_7 t_7 = 254$. Since $4t_7 = 256 > 254$, $M_7 < 4$. Checking $M_7 = 0, 1, 2, 3$, it is easy to see that there do not exist M_6 and M_7 such that $M_6 t_6 + M_7 t_7 = 254$.

- $M_5 = 2$

Using the same reasoning as in the last case, $M_5 t_5 = 97.4$ forces $M_6 t_6 + M_7 t_7 = 302.4 - 97.4 = 205$. We can again check $M_6 = 0, 1, 2, 3$ to see that this equation has no solution.

- $M_5 = 3$

In this case, $M_5 t_5 = 146.1$. Hence $\sum_{i=5}^7 M_i t_i$ will be an integer plus 0.1. Therefore (3) cannot be satisfied. However, if we allow equality to hold in the lower bound in (3), then $M_6 t_6 + M_7 t_7 = 302.1 - 146.1 = 156$. A simple check shows that the only solution is $M_6 = 3$, $M_7 = 0$. This corresponds to the loading $\{N^1, N^2\}$.

for which, in fact, $N_5^1 + N_5^2 = 3$, $M_6^1 + N_6^2 = 3$ and $N_7^1 + N_7^2 = 0$.

- $M_5 = 4$

Here $M_5 t_5 = 194.8$. Hence $\sum_{i=5}^7 M_i t_i$ will be an integer plus 0.8; clearly it cannot then satisfy (3).

- $M_5 = 5$

Then $M_5 t_5 = 243.5$. Hence $M_6 t_6 + M_7 t_7 = 302.5 - 243.5 = 59$. There is no integral linear combination of $t_6 = 52$ and $t_7 = 64$ equal to 59.

- $M_5 = 6$

Now $M_5 t_5 = 292.2$. Clearly adding even one to t_6 or t_7 will produce a number greater than 302.7; hence there are no solutions for $M_5 = 6$.

We have thus shown that (3) has no solutions, although $(M_5, M_6, M_7) = (3, 3, 0)$ produces a sum of 302.1, the value achieved by N^1 and N^2 . It follows that 302.1 is the maximum value of $\sum_{i=5}^7 M_i t_i$, so $302.1 + 1737.3 = 2039.4$ is the maximum for $N^1 \cdot t + N^2 \cdot t$. $\square$

Theorem 1 refers to the interpretation of the problem in which the local trucking constraint is held to apply to the total loading of both flatcars. The other interpretation—that this constraint applies only to each flatcar individually—is obviously looser, in that any loading confirming to the former interpretation will conform to the latter. Hence we may be able to load the flatcars with even less wasted space. In fact, we can pack them perfectly.

Theorem 2. There exists a loading $\{N^1, N^2\}$ satisfying constraints C1–C3 and (2) such that $\{N^1, N^2\}$ uses all available space.

Proof: Let

$$N^1 = (6 \ 2 \ 6 \ 0 \ 0 \ 0 \ 4) \quad \text{and} \quad N^2 = (0 \ 5 \ 2 \ 5 \ 2 \ 1 \ 2).$$

This loading satisfies constraints C1 – C3 and (2).

1. We have $N^1 + N^2 = (6 \ 7 \ 8 \ 5 \ 2 \ 1 \ 6)$. Hence $N_i^1 + N_i^2 \leq n_i$, so the first constraint is satisfied.
2. We compute $N^1 \cdot t = 1020$ and $N^2 \cdot t = 1020$; therefore the N^i satisfy the second constraint and pack the two flatcars perfectly.
3. The total weight on the first flatcar is $N^1 \cdot w = 28 < 40$ and on the second is $N^2 \cdot w = 31.5$. Hence the third constraint is satisfied.
4. Here $N^1 \cdot t' = 256 < 302$ and $N^2 \cdot t' = 277.4 < 302.7$. Therefore we satisfy (2) (but not, of course, (1)).

This verification, along with the fact observed that these represent a perfect loading of the flatcars, completes the proof. $\square$

3. Search for Optimal and Near-Optimal Solutions

In this section, we describe a computer program that can search for solutions to the problem under both interpretations of the local trucking constraint.

We first simplify the problem by replacing the two flatcars by a single flatcar with composite constraints. That is, it has 20.4 m of space available and can hold up to 80 metric tons. Under the more restrictive interpretation of the local trucking constraint, this single flatcar may only hold crates of types 5, 6 and 7 having thicknesses totaling 302.7 cm. Under the less-restrictive constraint, the car may hold up to 605.4 cm of these crates.

Once the program has found near optimal loadings of the composite flatcar (that is, ones that nearly fill it), it then tries to balance these loadings by distributing the crates between the two real flatcars, subject to the constraints of the problem. Clearly, any admissible loading of the two flatcars produces an admissible loading of the composite car. Hence this technique must, in fact, find the optimal loadings. It will be fairly efficient if the optimal or near-optimal loadings of the composite single car can be balanced; this in fact happens.

The program appears in Appendix B [omitted here]. Its structure reflects the division of the problem into composite loading and balancing described above. The main program checks all possible loadings of the composite car. It prints and attempts to balance those that come within user-specified limits of perfectly packing the composite car. All ways of balancing a loading of the composite car between the two flatcars are printed out.

Appendix A contains two sample runs of the program. The first shows the 60 ways of balancing the unique composite loading that achieve 0.6 cm of wasted space. (Note that the proof of **Theorem 1** shows that there is a unique composite loading with this wastage.) The second shows the six perfect loadings of two flatcars corresponding to ten loadings of the composite flatcar. (Four of the composite loadings cannot be balanced; the other six each balance in exactly one way.)

This method of solution has important advantages and disadvantages. By supplying the appropriate parameters to the program, one can obtain packings that waste less than a specified amount of space more than the minimum. (For example, there are exactly 1282 balancings that obey the more restrictive local trucking constraint and waste no more than 10 cm.) This can be crucial for real-world

applications. It is likely, for example, that a perfect packing is unacceptable in practice; a practical packing would need to have some "play."

Further, the given program is easily customized to solve closely related problems. If, for example, the measurement of one of the crates' weight or thickness is inaccurate, only two arrays within the program need be modified. Even a large change, like another trucking constraint, could be accommodated without major changes. A few changes would allow the program to optimize the for other quantities, for instance, a weighted sum of used space and weight. In fact, as it stands, the exhaustive results allow selection among optimal solutions on other criteria. The perfect loading presented in the solution to **Theorem 2**, for instance, has the greatest weight of the six perfect loadings, as a quick examination of Appendix A shows.

However, the program has a major problem that is crippling from a real-world point of view. Since it essentially uses an exhaustive search, it could not handle much greater numbers of crates. This algorithmic naïveté means that the program applies only in relatively simple situations. In the next section, we consider this weakness in greater detail.

4. Critique of Method and Generalization of Problem

The program described in the preceding section and the proofs of Section 2 depend strongly on the specifics of the given problem. In particular, the algorithm for finding optimal loadings becomes impractically slow as the number of crates increases. It might be thought that that is simply the fault of this algorithm and that another might do much better. However, we show in this section that the fact the problem of finding optimal loadings is hard in a formal sense; it is NP-complete. At the end of this section, we suggest a heuristic algorithm to search for solutions to the hard general problem.

A problem P is said to be in the class NP if there is a nondeterministic Turing machine algorithm that can solve P in a number of steps bounded by a polynomial in the size of the input [Garey and Johnson 1979]. (More informally, a problem is in NP if there exists a Turing-machine algorithm that can check in polynomial time a solution it receives from an oracle.) Certain problems in NP are called NP-complete. The existence of a polynomial-time deterministic Turing machine algorithm for any one NP-complete problem implies

the existence of a polynomial-time deterministic Turing machine algorithm for every problem in NP. S.A. Cook proved the existence of NP-complete problems in 1971 [Cook 1971].

Theorem 3. The problem LOADING FLATCARS is NP-complete, where LOADING FLATCARS is defined as follows:

Given a set N of boxes, and for each $n \in N$, a weight $w(n) \in Z^+$ and a thickness $t(n) \in Z^+$, together with four positive integers k , W , T and T' , do there exist k disjoint subsets $S_i \subset N$ such that, for $i = 1, \dots, k$, the following hold:

$$\sum_{n \in S_i} w(n) \leq W,$$

$$\sum_{n \in S_i} t(n) \leq T,$$

$$\sum_{i=1}^k \sum_{n \in S_i} t(n) \geq T'?$$

[Note that here $w(n)$ represents the weight of the box n , $t(n)$ the thickness of the box n , and the S_i the flatcars. Although thicknesses and weights need not be integers, any measurements made of them will be rational because of the finite precision of measuring instruments. We can then multiply to clear denominators and obtain an integer-based problem. The first two constraints represent the size and weight limits of individual flatcars; the third makes possible maximization of thickness (i.e. minimization of wasted floor space).]

Proof: We must first show that LOADING FLATCARS is in NP. If the oracle provides a collection of k disjoint subsets $\{S_i\}$, we can quickly check whether $\{S_i\}$ is a solution to the problem. To see that LOADING FLATCARS is NP-complete, we note that restricting it to the cases where $k = 1$ and ignoring constraint 2. reduces LOADING FLATCARS to KNAPSACK:

Given a finite set N and, for each $n \in N$, a weight $w(n) \in Z^+$ and a thickness $t(n) \in Z^+$, together with positive integers W and T' , does there exist a subset $S \subset N$ such that $\sum_{n \in S} w(n) \leq W$ and $\sum_{n \in S} t(n) \geq T'$?

KNAPSACK, as defined above, is an NP-complete problem [Karp 1972]. Thus, any polynomial-time algorithm for LOADING FLATCARS will also be a polynomial-time algorithm for KNAPSACK. Since KNAPSACK is NP-complete, LOADING FLATCARS is harder than KNAPSACK, and LOADING FLATCARS is NP-complete.

It should be noted that the given problem had the additional local trucking constraint, which was vital to our proof of Theorem 1. Although in this case the added condition made the problem at hand

more tractable, general constraints of the same kind make the task of finding a general solution to the general problem more difficult. Any polynomial-time algorithm that solved the problem with such added constraints would solve it without them.

Also, notice that the algorithm described in the preceding section involved breaking the problem into two parts. First we loaded a single large flatcar subject to constraints on total thickness and weight, trying to maximize used space. This, of course, includes an instance of KNAPSACK (with some added constraints). Then we tried to distribute this loading between the two smaller flatcars. This includes the general problem PARTITION, also known to be NP-complete [Karp 1972]:

Given a finite set A and, for each $a \in A$, a size $s(a) \in E^+$, is there a subset $A' \subset A$ such that:

$$\sum_{a \in A'} s(a) = \sum_{a \in A - A'} s(a)?$$

It might seem that converting a single NP-complete problem into two NP-complete problems is no great improvement. However, for problems of small size this is, in fact, more efficient, because the KNAPSACK problem is less complicated in practice than LOADING FLATCARS and we need only solve PARTITION for solutions of KNAPSACK that are near-optimal.

Had we been given a more general instance of the problem, we would not have been able to solve it as we did, both in Theorem 1, where we exploited the local trucking constraint to prove an optimal solution, and in the computer program, where we conducted an exhaustive search of a slightly smaller problem. In a general problem, of loadings satisfying constraints like C1-C3 and equations (1) and (2) of the original problem form a convex set L in Z^n for some large n . The loading that minimizes wasted floor space will lie near the boundary of L . To search for this loading, we could choose a random loading in L and then add boxes until we reached the boundary. Once we were at the boundary, we could take a random walk over it by making exchanges of boxes and adding boxes. By iterating this process, we are likely to find a loading with near-minimum wastage.

5. Conclusion

We solved the given problem exactly under two reasonable interpretations of the local trucking constraint and proved our solutions optimal, which had 0.6 cm wasted for one interpretation

"... we proved
that the
general
problem is
hard...."

and zero wasted for the other. Since these solutions are, of course, particular to the exact problem given, we also included a computer program that could easily be altered to optimize for any quantity (for example, weight rather than used space) among loadings satisfying any reasonable constraints, as long as the total size of the problem remains tractable. The program can also produce suboptimal loadings, to provide room for measurement error. Finally, we proved that the general problem is, in a well-defined sense, hard: No known deterministic algorithm can solve it in polynomial time.

References

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Appendix A. Computational Results

This appendix contains two runs of the program. The first run shows the 60 solutions to the problem with the more-restrictive local trucking constraint which waste 0.6 cm. The second run shows the 6 perfect loadings to the problem with the less-restrictive local trucking constraint, along with all the loadings of a composite car having twice the capacity of each given flatcar which waste 0.5 cm or less. Each row of numbers in the left-hand column represents a loading of the composite car, most of which could be broken into loadings for the two flatcars. Each pair of rows of numbers in the right-hand column represents a loading of the two flatcars.

Program to search for optimal flatcar loadings

Which problem?

(1 = restrictive local trucking constraint; 302.7 cm total)

(2 = generous constraint; 302.7 cm on each car)

Print sumloadings up to how much wasted space?
(in units of 0.1 cm)

6

Balance sumloadings up to how much wasted space?

6

8 7 9 6 3 3 0 2039.4

Crane

1 2 3 4 5 6 7

Total
Weight

Used
Space

6 2 9 1 0 0 0

27.5

1019.9

2 5 0 5 3 3 0

39.5

1019.5

Crate							Total	Used
1	2	3	4	5	6	7	Weight	Space
4	7	4	3	0	0	0	34.5	1020.0
4	0	5	3	3	3	0	32.5	1019.4
7	3	5	3	0	0	0	29.5	1019.4
1	4	4	3	3	3	0	37.5	1020.0
0	7	6	4	0	0	0	29.0	1019.8
8	0	3	2	3	3	0	38.0	1019.6
5	2	9	1	1	0	0	29.5	1019.9
3	5	0	5	2	3	0	37.5	1019.5
3	7	4	3	1	0	0	36.5	1020.0
5	0	5	3	2	3	0	30.5	1019.4
6	3	5	3	1	0	0	31.5	1019.4
2	4	4	3	2	3	0	35.5	1020.0
4	2	9	1	2	0	0	31.5	1019.9
4	5	0	5	1	3	0	35.5	1019.5
2	7	4	3	2	0	0	38.5	1020.0
6	0	5	3	1	3	0	28.5	1019.4
5	3	5	3	2	0	0	33.5	1019.4
3	4	4	3	1	3	0	33.5	1020.0
3	2	9	1	3	0	0	33.5	1019.9
5	5	0	5	0	3	0	33.5	1019.5
4	3	5	3	3	0	0	35.5	1019.4
4	4	4	3	0	3	0	31.5	1020.0
8	1	0	6	3	0	0	34.0	1019.7
0	6	9	0	0	3	0	33.0	1019.7
4	6	4	3	0	1	0	33.5	1020.0
4	1	5	3	3	2	0	33.5	1019.4
7	2	5	3	0	1	0	28.5	1019.4
1	5	4	3	3	2	0	38.5	1020.0
0	6	6	4	0	1	0	28.0	1019.8
8	1	3	2	3	2	0	39.0	1019.6
5	7	0	5	0	1	0	35.5	1019.5
3	0	9	1	3	2	0	31.5	1019.9
5	1	9	1	1	1	0	28.5	1019.9
3	6	0	5	2	2	0	38.5	1019.5

Crate 1 2 3 4 5 6 7							Total Weight	Used Space
3	6	4	3	1	1	0	35.5	1020.0
5	1	5	3	2	2	0	31.5	1019.4
6	2	5	3	1	1	0	30.5	1019.4
2	5	4	3	2	2	0	36.5	1020.0
4	7	0	5	1	1	0	37.5	1019.5
4	0	9	1	2	2	0	29.5	1019.9
4	1	9	1	2	1	0	30.5	1019.9
4	6	0	5	1	2	0	36.5	1019.5
2	6	4	3	2	1	0	37.5	1020.0
6	1	5	3	1	2	0	29.5	1019.4
5	2	5	3	2	1	0	32.5	1019.4
3	5	4	3	1	2	0	34.5	1020.0
3	7	0	5	2	1	0	39.5	1019.5
5	0	9	1	1	2	0	27.5	1019.9
3	1	9	1	3	1	0	32.5	1019.9
5	6	0	5	0	2	0	34.5	1019.5
8	2	3	2	3	1	0	40.0	1019.6
0	5	6	4	0	2	0	27.0	1019.8
1	6	4	3	3	1	0	39.5	1020.0
7	1	5	3	0	2	0	27.5	1019.4
4	2	5	3	3	1	0	34.5	1019.4
4	5	4	3	0	2	0	32.5	1020.0
8	0	0	6	3	1	0	33.0	1019.7
0	7	9	0	0	2	0	34.0	1019.7
0	7	9	0	0	2	0	34.0	1019.7
8	0	0	6	3	1	0	33.0	1019.7
4	5	4	3	0	2	0	32.5	1020.0
4	2	5	3	3	1	0	34.5	1019.4
7	1	5	3	0	2	0	27.5	1019.4
1	6	4	3	3	1	0	39.5	1020.0
0	5	6	4	0	2	0	27.0	1019.8
8	2	3	2	3	1	0	40.0	1019.6
5	6	0	5	0	2	0	34.5	1019.5
3	1	9	1	3	1	0	32.5	1019.9

Crate							Total Weight	Used Space
1	2	3	4	5	6	7		
5	0	9	1	1	2	0	27.5	1019.9
3	7	0	5	2	1	0	39.5	1019.5
3	5	4	3	1	2	0	34.5	1020.0
5	2	5	3	2	1	0	32.5	1019.4
6	1	5	3	1	2	0	29.5	1019.4
2	6	4	3	2	1	0	37.5	1020.0
4	6	0	5	1	2	0	36.5	1019.5
4	1	9	1	2	1	0	30.5	1019.9
4	0	9	1	2	2	0	29.5	1019.9
4	7	0	5	1	1	0	37.5	1019.5
2	5	4	3	2	2	0	36.5	1020.0
6	2	5	3	1	1	0	30.5	1019.4
5	1	5	3	2	2	0	31.5	1019.4
3	6	4	3	1	1	0	35.5	1020.0
3	6	0	5	2	2	0	38.5	1019.5
5	1	9	1	1	1	0	28.5	1019.9
3	0	9	1	3	2	0	31.5	1019.9
5	7	0	5	0	1	0	35.5	1019.5
8	1	3	2	3	2	0	39.0	1019.6
0	6	6	4	0	1	0	28.0	1019.8
1	5	4	3	3	2	0	38.5	1020.0
7	2	5	3	0	1	0	28.5	1019.4
4	1	5	3	3	2	0	33.5	1019.4
4	6	4	3	0	1	0	33.5	1020.0
0	6	9	0	0	3	0	33.0	1019.7
8	1	0	6	3	0	0	34.0	1019.7
4	4	4	3	0	3	0	31.5	1020.0
4	3	5	3	3	0	0	35.5	1019.4
5	5	0	5	0	3	0	33.5	1019.5
3	2	9	1	3	0	0	33.5	1019.9
3	4	4	3	1	3	0	33.5	1020.0
5	3	5	3	2	0	0	33.5	1019.4
6	0	5	3	1	3	0	28.5	1019.4
2	7	4	3	2	0	0	38.5	1020.0

Crate							Total	Used
1	2	3	4	5	6	7	Weight	Space
4	5	0	5	1	3	0	35.5	1019.5
4	2	9	1	2	0	0	31.5	1019.9
2	4	4	3	2	3	0	35.5	1020.0
6	3	5	3	1	0	0	31.5	1019.4
5	0	5	3	2	3	0	30.5	1019.4
3	7	4	3	1	0	0	36.5	1020.0
3	5	0	5	2	3	0	37.5	1019.5
5	2	9	1	1	0	0	29.5	1019.9
8	0	3	2	3	3	0	38.0	1019.6
0	7	6	4	0	0	0	29.0	1019.8
1	4	4	3	3	3	0	37.5	1020.0
7	3	5	3	0	0	0	29.5	1019.4
4	0	5	3	3	3	0	32.5	1019.4
4	7	4	3	0	0	0	34.5	1020.0
2	5	0	5	3	3	0	39.5	1019.5
6	2	9	1	0	0	0	27.5	1019.0

total balancings: 60

Program to search for optimal flatcar loadings

Which problem?

(1 = restrictive local trucking constraint; 302.7 cm total)

(2 = generous constraint; 302.7 cm on each car)

2

Print sumloadings up to how much wasted space?

(in units of 0.1 cm)

5

Balance sumloadings up to how much wasted space?

0

2 7 9 6 4 4 3 2039.9

3 7 9 6 3 4 3 2039.9

4 7 9 6 2 4 3 2039.9

5 7 8 5 3 1 6 2040.0

cannot balance

5 7 9 6 1 4 3 2039.9

6 6 8 5 2 2 6 2040.0

cannot balance

6 7 8 4 2 1 6 2040.0

Crate							Total Weight	Used Space
1	2	3	4	5	6	7		
0	5	2	5	2	1	2	31.5	1020.
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
6	7	9	6	0	4	3	2039.9	
7	5	8	5	1	3	6	2040.0	
cannot balance								
7	5	8	5	6	2	3	2039.5	
7	6	8	5	1	2	6	2040.0	
1	4	2	5	1	2	2	28.5	1020.0
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
7	6	8	5	6	1	3	2039.5	
7	7	8	5	1	1	6	2040.0	
1	5	2	5	1	1	2	29.5	1020.0
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
7	7	8	5	6	0	3	2039.5	
8	4	8	5	0	4	6	2040.0	
cannot balance								
8	4	8	5	5	3	3	2039.5	
8	5	8	5	0	3	6	2040.0	
2	3	2	5	0	3	2	25.5	1020.0
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
8	5	8	5	5	2	3	2039.5	
8	6	8	5	0	2	6	2040.0	
2	4	2	5	0	2	2	26.5	1020.0
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
8	6	8	5	5	1	3	2039.5	
8	7	6	5	3	3	4	2039.5	
8	7	8	3	5	4	2	2039.5	
8	7	8	5	0	1	6	2040.0	
2	5	2	5	0	1	2	27.5	1020.0
6	2	6	0	0	0	4	28.0	1020.0
total balancings: 1								
8	7	8	5	5	0	3	2039.5	

Appendix B. Program Listing [omitted]



1. Introduction

When determining the loading pattern for units of various thickness and weight on railroad flatcars, the problem is mathematically stated as follows. The given problem is concerned with